

Noël Margaret Clark

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Education

Ph.D. student: Fish & Wildlife Biology	GPA: 4.00
University of Montana, Missoula, MT	<i>expected 5.28</i>
NSF Graduate Research Fellowship – 2023-2028	
M.S.: Fish, Wildlife, & Conservation Biology	GPA: 4.00
Colorado State University, Fort Collins, CO	5.25
B.S.: Marine Science , Minor: Statistics	GPA: 3.76
California Polytechnic State University, San Luis Obispo, CA	06.21
University Honors Program, Magna Cum Laude, President's List, Dean's List	

Awards & Honors

10.25	Graduate Degree Program in Ecology Small Research Grant – Colorado State University (recommended)
09.25	Colorado/Wyoming American Fisheries Society Scholarship – Colorado State University
08.25	Kurt Fausch - Shigeru Nakano Scholarship in Fisheries and Aquatic Sciences – Colorado State University
05.25	Eugene Maughan Memorial Scholarship – Western Division American Fisheries Society
10.24	West Denver Chapter of Trout Unlimited Scholarship – Colorado State University
03.24	Best Student Poster – CO/WY American Fisheries Society Chapter Annual Meeting
03.24	American Fisheries Society Student Travel Grant – CO/WY American Fisheries Society Chapter
06.23	Fred Rasmussen Memorial Scholarship – Collegiate Peaks Chapter Trout Unlimited
04.23	Robert J. Behnke Rocky Mountain Flycasters Research Fellowship – Colorado State University
04.23	Summer Institute in Statistical Genetics Scholarship – University of Washington
03.23	Graduate Research Fellowship Program – National Science Foundation
08.22	Mentee at Footprints Running – Patagonia Trail Running & Protect Our Winters
11.20	Honorable Mention for Best Student Talk – Pacific Coast Shellfish Growers Association Conference
06.20 ^t	Medtronic Dependent Scholarship – Medtronic Inc. <i>*also awarded in 2017 & 2019</i>
12.19	Brickhouse Security College Scholarship – Brickhouse Security
06.19	Undergraduate Research Scholar Award – Bill and Melinda Frost Fund

Research Experience

Ph.D. Dissertation; Franke College of Forestry and Conservation, University of Montana, Missoula, MT	01.23-Pres
- Contributing to a 6-year mark-recapture field study of masu salmon (<i>Oncorhynchus masou</i>) in Hokkaido, Japan in collaboration with the Hokkaido University Tomakomai Experimental Forest Research Station - Designing, implementing, and managing a manipulative habitat restoration before- after-control-impact experiment, including installation of 160 log structures harvested from the riparian area into 5.3 km of stream - Leading field surveys to monitor experimental installations and associated geomorphic process changes through physical habitat measurements, including Wolman's pebble count transects, microhabitat assessments, leaf litter surveys, and macroinvertebrate production via Surber sampling - Collaborating with faculty, researchers, graduate students, and technicians of the Hokkaido University, Japan - Began dissertation work at Colorado State University, Fort Collins, CO before transferring to University of Montana, Missoula, MT with my advisor, Dr. Yoichiro Kanno	
M.S. Assistantship; Warner College of Natural Resources, Colorado State University, Fort Collins, CO	01.23-05.25
- Investigated evolutionary history of Cutthroat Trout (<i>Oncorhynchus</i> spp.) lineages using next generation sequencing techniques to inform coldwater fisheries conservation management by Colorado Parks and Wildlife - Employed population genetic theory and genomic demographic reconstruction techniques to further elucidate the history of Cutthroat Trout lineage diversification and unrecorded human stocking in Colorado - Wrote, managed, and maintained a genomics pipeline for cleaning, trimming, mapping, and joint genotype calling whole genome sequence data for Cutthroat Trout samples - Participated in field sampling for stable isotope and diet composition studies on brown trout (<i>Salmo trutta</i>) and mottled sculpin (<i>Cottus bairdii</i>) in the Blue River of the Colorado Rocky Mountains	

- Corpsmember;** NOAA Southwest Fisheries Science Center, Watershed Stewards Program (WSP) 10.21-08.22
- Engaged in watershed research and civic service through a special program of the California Conservation Corps
 - Contributed to long-term research and monitoring of ESA-listed Central Coast coho salmon (*O. kisutch*) and threatened steelhead trout (*O. mykiss*) on Scott Creek, CA, through the NOAA SWFSC Scott Creek team
 - Maintained a weir, adult, and juvenile out-migrant traps on Scott Creek, with handling and species ID of salmonids
 - Organized 3 community volunteer events for trail building, invasive weed removal, and native plant restoration
 - Taught five 90-minute watershed science lessons in a local Title-I elementary school
 - Tagged fish with passive-integrated-transponder (PIT) and t-bar tags, and extracted DNA of salmonids
 - Identified salmonid redds, live fish, and carcasses during spawning ground surveys across watersheds
 - Processed carcasses, including in-field head removal and lab-based otolith and coded wire tag (CWT) extraction
 - Operated handheld GPS data collection device, PIT tag reader (HPR), and remote data loggers
 - Built, tuned, and maintained swim-through duplex looped wire PIT tag antennas
- Research Technician;** Ruttenberg Marine Conservation Lab, Cal Poly SLO, CA 01.18-10.21
- Monitored populations of Pismo clams (*T. stultorum*) through annual statewide and local monthly surveys
 - Trained new graduate master's student and new members on laboratory and field protocols as team leader
 - Conducted intertidal field work and collecting quality data in unfavorable conditions, including at night
 - Awarded research stipend of \$3,000 to conduct an independent project on growth rates of Pismo clams
 - Developed research scope, clam aging methods, and statistical analyses through literature review
 - Presented research findings through multiple symposiums and conferences
 - Performed condition indices, histological microtoming, and chlorophyll sampling
- Research Technician;** D.E. Wendt Lab, Cal Poly SLO, CA 12.19-10.21
- Conducted various research projects with the California Collaborative Research Program (CCFRP), Office of Naval Research, and stock assessments with the National Marine Fisheries Service
 - Taxonomic ID of live marine organisms, including invertebrates and rockfishes (*Sebastes spp.*)
 - Implanted t-bar and PIT tags, otolith extraction, measuring, and handling
 - Operated a CTD, Secchi disk, fish descending device, and handheld GPS during fishing vessel drift periods
 - Entered, managed, and verified data within long term, cross institutional datasets
 - Created and maintained a Microsoft Access database, including QA/QC using ArcMap, R/R.Studio, and JMP
- Lab Aide;** Bodega Ocean Acidification Lab, UC Davis Bodega Marine Lab, Bodega Bay, CA 04.16-09.18
- Aided Ph.D. candidates with lab and field research for theses
 - Deployed controlled salinity gradient traps for invasive species research in Tomales Bay, CA
 - Handled critically endangered White Abalone (*H. sorenseni*) in desiccating ovens and on micro-scales
 - Cleaned and maintained wet lab and aquaculture set-ups, including for temperature pH/controlled experiments

Publications

- Kanno, Y., **N. Clark**, K.C. Pregler, S. Kim. 2025. Integrated analysis of marked and count data to characterize fine-scale stream fish movement. *Oecologia* 207:25.
- Hosoki, T.K., **N. Clark**, R. Futamura, S. Moriyama, O. Kishida, Y. Kanno. 2024. A comparison of sex-specific markers for two wild masu salmon populations in Hokkaido, Japan. *Aquaculture, Fish, and Fisheries* 4:e194.
- Marquardt, A., S. Park, E. Maietta, **N. Clark**, G. Waltz, L. Needles, B. Ruttenberg. 2023. Status of intertidal populations of Pismo clams (*Tivela stultorum*) across their range in California. *Reg. Studies in Marine Science* 65:103063.
- Marquardt, A., **N. Clark**, E. Maietta, S. Park, B. Ruttenberg. 2022. Reproduction, body condition, age, and growth of a large sandy intertidal bivalve, *Tivela stultorum*, Mawe 1823. *Aquatic Biology* 31:19-30.
- Clark, N.**, E. Maietta, A. Marquardt, B. Ruttenberg. 2021. Age and Growth Determination of the Pismo Clam (*Tivela stultorum*) in California. Cal Poly Biological Sciences Department, senior thesis. Creative Commons.

Publications in Preparation

- Clark, N.**, K. Rogers, E. Anderson, W.C. Funk, Y. Kanno, C. Wells. Whole Nuclear Genomes Identify Novel Relationships of Cutthroat Trout in the Southern Rocky Mountains.
- Kanno, Y., **N. Clark**, T.J. Knutson, K.D. Fausch, M.B. Hooten. Characterizing Stream Fish Movement Distances Using Mark-recapture Methods: Perspectives and a New Unified Statistical Approach.
- Paul, B., **N. Clark**, K. Pregler, S. Kim, Y. Kanno. Sub-seasonal Correlations in Survival Between Body Size Classes and Population Dynamics of Non-game Fishes in Coolwater Streams.

Contributed Presentations

- Clark, N.,** K. Rogers, E. Anderson, W.C. Funk, Y. Kanno, C. Wells. (2025, November). Using DNA to investigate Cutthroat Trout diversity in Colorado. West Denver Chapter of Trout Unlimited Meeting, Denver, CO. Invited oral presentation.
- Clark, N.,** K. Rogers, E. Anderson, W.C. Funk, Y. Kanno, C. Wells. (2025, May). Whole genome sequencing to characterize Cutthroat Trout populations across the Continental Divide. Western Division American Fisheries Society Annual Meeting, Denver, CO. Oral presentation.
- Clark, N.,** K. Rogers, E. Anderson, W.C. Funk, Y. Kanno, C. Wells. (2025, May). Whole genome sequencing to characterize Cutthroat Trout populations across the Continental Divide. Greenback Cutthroat Trout Recovery Team Meeting, Denver, CO. Invited oral presentation.
- Clark, N.,** K. Rogers, E. Anderson, W.C. Funk, Y. Kanno, C. Wells. (2025, March). Whole genome sequencing to characterize Cutthroat Trout populations across the Continental Divide. CSU Department of Fish, Wildlife, & Conservation Biology Seminar Series, Fort Collins, CO. Oral presentation.
- Clark, N.,** K. Rogers, E. Anderson, W.C. Funk, Y. Kanno, C. Wells. (2024, December). Characterizing Cutthroat Trout (*Oncorhynchus clarkii*) populations across the Continental Divide. Colorado River Cutthroat Trout Recovery Team Meeting, Grand Junction, CO. Oral presentation.
- Clark, N.,** K. Rogers, E. Anderson, W.C. Funk, Y. Kanno, C. Wells. (2024, September). Whole genome sequencing to characterize Colorado River Cutthroat Trout (*Oncorhynchus clarkii pleuriticus*) populations across the Continental Divide. National Annual American Fisheries Society Meeting, Honolulu, HI. Oral presentation.
- Clark, N.,** K. Rogers, Y. Kanno, W.C. Funk, C. Wells. (2024, March). Whole genome sequencing to characterize Colorado River Cutthroat Trout (*Oncorhynchus clarkii pleuriticus*) populations across the Continental Divide. Colorado-Wyoming Chapter American Fisheries Society Meeting, Laramie, WY. Poster.
- Clark, N.,** I. Tang, R. Bond, A. Hay, C. Kern, J. Meko, J. Perez, J. Kiernan. (2023, October). Novel Chinook salmon spawning observations in the Santa Cruz Mountains, California. Hokkaido University School of Agriculture and the Salmon Science Society, Salmon Science Research Group Special Seminar. Invited oral presentation.
- Clark, N.,** I. Tang, R. Bond, A. Hay, C. Kern, J. Meko, J. Perez, J. Kiernan. (2022, April). Novel Chinook salmon spawning observations in the Santa Cruz Mountains. Salmonid Restoration Federation, Santa Cruz, CA. Poster.
- Clark, N.** (2021, January). Pismo clam (*Tivela stultorum*) growth determination using the Von Bertalanffy Growth Function. Cal Poly College of Science and Math, MSCI 439 Fisheries Science and Resource Management. Virtual class lecture.
- Clark, N.,** A. Marquardt, E. Maietta, B. Ruttenberg. (2020, November). Age and growth determination of the Pismo clam (*Tivela stultorum*) in California. Western Society of Naturalists, virtual conference. Speed talk.
- Clark, N.,** A. Marquardt, E. Maietta, B. Ruttenberg. (2020, October). Age and growth determination of the Pismo clam (*Tivela stultorum*) in California. Pacific Coast Shellfish Growers Association, virtual conference. Oral presentation.
- Clark, N.,** A. Marquardt, E. Maietta, B. Ruttenberg. (2019, October). Age and growth determination of the Pismo clam (*Tivela stultorum*) in California. Western Society of Naturalists, Ensenada, B.C. Mexico. Poster.
- Clark, N.,** A. Marquardt, E. Maietta, B. Ruttenberg. (2019, July). Age and growth determination of the Pismo clam (*Tivela stultorum*). Frost Undergraduate Research Symposium, Cal Poly San Luis Obispo. Oral presentation.

TEACHING EXPERIENCE

- Instructor;** Colorado State University, Graduate Degree Program in Ecology (GDPE), Fort Collins, CO 01.26-03.26
- Co-instructor for 10 graduate students in ECOL592, a seminar style science communication course developed by myself, Maricela Alaniz (GDPE PhD student), and Dr. Jamie Dahl (Assistant Professor in the Department of Human Dimensions of Natural Resources, Warner College of Natural Resources, CSU)
 - Covering a wide range of science communication topics including principles of design, constructing graphical abstracts and conceptual figures, K-12 science education, engaging with media, science and poetry, and scientific presentation skills
- Teaching Assistant;** Colorado State University, Warner College of Natural Resources, Fort Collins, CO 01.25-05.25
- Primary instructor for 40 students across two sections of FW 301 – Ichthyology
 - Presented lectures on fish diversity, taxonomy, aquatic ecology, and morphometrics
 - Lead laboratory activities with preserved specimen from the CSU Ichthyology collection
 - Established assignments, created quiz and exam questions, and graded all course submissions

Teaching Assistant; Colorado State University, Warner College of Natural Resources, Fort Collins, CO	01.23-05.23
• <i>TA for 100 students in FW 104 – Wildlife Ecology and Conservation (entire semester)</i> • Presented lectures on previous research projects and on the principles of marine conservation • Established assignments, created quiz and exam questions, and graded all course submissions • <i>TA for 20 students in FW 375 – Field Wildlife Studies over Spring break (03.10.2023-03.18.2023)</i> • Transported students in 12-passenger van 2,400 miles across the Southwest on major highways and 4WD roads • Participated in field wildlife studies lectures, demonstrations, and restoration work with individuals from AZ Game & Fish, Ted Turner's Ladder Ranch, the National Museum of Natural History Southwestern Research Station, AZ State University, The Nature Conservancy, and Borderland Plants Nonprofit • Conducted volunteer riparian restoration and native plant nursery work in Patagonia, AZ • Participated in an acoustic monitoring survey for elk, mountain lion, and ESA-listed Mexican Gray Wolves • Exposed to herptiles research, monitoring, and conservation techniques	
Teaching Assistant; Cal Poly College of Science and Math, San Luis Obispo, CA	12.20-03.21
• TA for the Fisheries Science and Resource Management (MSCI 439) class • Presented lecture material on growth and age modeling using the Von Bertalanffy Growth Function • Established online coursework, maintained online assignments, created quizzes, and graded exams	
STAT 511 Grader; Cal Poly Statistics Department, San Luis Obispo, CA	09.20-12.20
• Graded a graduate biology student introductory statistics course • Worked with the statistics professor to return feedback to students and grade labs fairly across multiple sections	

Scientific Communication

Riparian Education: Created and presented two interactive lessons to two 4 th grade classes - Lakewood, CO	04.24
Salmon Science Society Special Seminar: Invited oral presentation - University of Hokkaido, Japan	10.23
World Fish Migration Day: Presented a poster from the SRF conference to a public audience - Santa Cruz, CA	05.22
Watershed Awareness Volunteer Event: Organized 3 volunteer based invasive plant removals - Santa Cruz, CA	06.22-07.22
Wonders of Watersheds! Series: Taught watershed curriculum to two Title-I 5 th grade classes - Santa Cruz, CA	01.22-02.22
CCFRP Volunteer Angler Appreciation BBQ: Created a data summary presentation for an informal audience	04.20
Cal Poly Pier Open House: Cal Poly Center for Coastal Marine Science - Avila, CA	11.19
Undergraduate Research Panelist: Q&A on research opportunities for marine science students - Cal Poly	11.19
Pismo Beach Clam Festival: Outreach booth on Pismo clam life history and ongoing research - Pismo Beach, CA	10.19
Santa Barbara CC Field Studies Class: Trained students in intertidal field sampling protocols - Morro Bay, CA	04.18
Docent at Central Coast Aquarium: Facilitated encounters and narrated feedings to the public - Avila, CA	10.17-03.18

Service, Professional Development, and Certifications

Manuscript Reviewer; <i>Freshwater Biology, Animal Conservation</i>	2025-26
American Fisheries Society Student/Early Career Section; Anonymous application reviewer	2025-26
Integrated Population Modeling Workshop; Colorado State University	10.2025
Graduate Student Organization; President, vice president, faculty liaison	2024-26
ConGen 2023 Workshop; University of Montana	09.2023
Summer Institute in Statistical Genetics; University of Washington	06.2023
American Fisheries Society; National, Western Division, and CO/WY Chapter Student Member	2023-26
Wilderness First Aid & CPR and Swift Water Safety I; Sierra Mountain Rescue	10.2021
National Shellfisheries Society; Student Member	2020-21
Cal Poly Marine Sciences Club; Executive Board Member, Cal Poly San Luis Obispo	2020-21
Cal Poly Climbing Club & Team; Co-Founder and President, Cal Poly San Luis Obispo	2018-21
American Fisheries Society; National, Western Division, and CA/NV Chapter Student Member	2018-20
PADI Open Water Dive Certification; Professional Association of Diving Instructors	11.2018

Technical Skills

Field Work – fisheries electrofishing surveys, salmonid spawning ground surveys, intertidal field sampling, deploying in-situ research devices, sampling from commercial passenger fishing vessels (CPFVs), tagging organisms for mark-recapture studies, macroinvertebrate Surber sampling, performing invasive weed management, and repeatedly lifting up to 75lbs

Data Analysis – Bayesian hierarchical modeling, bioinformatics, processing genomic data, writing functioning and reproducible R code, determining statistical testing methods, QA/QC of long-term, cross institutional, and sensitive datasets, and developing standard operating procedure documents for data integrity and longevity

Laboratory Work – taxonomic identification and handling of live and dead fishes, extracting otoliths and CWT, PIT and t-bar tagging, fileting fish and shucking invertebrates using research protocol, maintenance of aquaculture and wet lab set-ups for finfishes and invertebrates, and operation of desiccating oven, high-powered microscope with ImageJ, and high-powered precision balance

Machinery – operating 4x4 manual transmission vehicles, hand tools, and loading and transporting boats on trailers

WORK EXPERIENCE

Front Desk; Boulder Rock Club, Boulder, CO 11.22-7.24

- Managing membership requests, checking-in patrons, and providing a friendly, welcoming atmosphere
- Proficiency with Rock Gym Pro (RGP) for memberships, data entry, POS, and calendar events
- Giving top rope and lead belay tests, enforcing safety rules and regulations
- Completing retail tasks like restocking inventory, selling items, and organizing rental equipment

Warehouse Associate; FEED Co-operative, Petaluma, CA 07.21-09.21

- Received, organized, and loaded fresh produce deliveries from small local farms for local restaurant wholesale and community sourced agriculture (CSA) boxes
- Lifted up to 75 lbs. of produce boxes repetitively
- Operated hydraulic equipment safely within an industrial warehouse

Climbing Park Supervisor; Cal Poly ASI, San Luis Obispo, CA 06.18-06.21

- Student Manager for a staff of 13 supervisors from 04.19-06.20
- Developed and implemented technical trainings and safety protocol needed to operate the climbing park facility
- Enforced policies to maintain a safe and welcoming environment, including COVID-19 restrictions

Front Desk; The Pad Climbing, San Luis Obispo, CA 09.18-06.21

- Provided introductory tours, ran customer service, and ensured safety of gym members
- Volunteered as a rock climbing belayer, welcomed new patrons, and answered questions
- Coordinated events with the Pad gym and the Cal Poly Climbing Club as president